
UNIFIED MECHANICS

Volume V

Formal Register, Methods, and Reproducibility

Joseph Shields

VOLUME V OF V

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A CORRIGIBLE EDITION. This text is corrigible. Earlier manifests and printings are retained as provenance, not as immutable authority. Corrections are recorded by version and supersession history; no scientific claim is frozen.

ABSTRACT

This volume is the audit layer: axioms and definitions, dependency graph, scientifically graded results, computational certificates, empirical protocols, supersession history, and the current frontier register. A concise source-disposition summary remains in the book; the complete 491-source crosswalk is supplied as a machine-readable audit supplement so that the scientific register remains readable without sacrificing provenance.

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The living scientific record

1.1 Purpose of the register

This volume is the audit surface for the series. It does not add another layer of physical ontology. It records the commitments on which the argument depends, the meanings of its terms, the scientific grade of each inherited theorem, the relation between source documents and rewritten chapters, and the conditions under which a result must be revised or abandoned.

A register is necessary because the corpus developed through books, technical papers, addenda, computational notebooks, generated tables, data files, manifests, and several editorial states. Later documents often corrected earlier ones, while exact duplicates appeared in different formats or directories. Preserving everything therefore cannot mean repeating every sentence. It means preserving each distinct claim, equation, dataset, counterexample, certificate, failure, and historical transition in a structure where its current disposition is visible.

1.2 No-freeze versioning

Scientific content is never frozen. At most, an analysis protocol may be pre-registered for the duration of a test. The living edition separates four operations:

Version

a citable state of the manuscript, code, data, and register;

Correction

a change that repairs an error without changing the intended claim;

Supersession

a new claim or derivation that replaces an earlier one while retaining its provenance;

Withdrawal

a route that no longer supports the stated conclusion, retained so that it cannot silently return.

No manifest may declare a proposition scientifically immutable. Reproduc-

tion means that a result can be regenerated from a declared historical state; it does not require future editions to continue believing the result.

EDITORIAL RULE 1.1 — PROVENANCE WITHOUT AUTHORITY

Every superseded or withdrawn item remains addressable by source path and digest. Its survival in the record establishes what was written and tested, not that the claim is true. The current scientific grade is governed by the latest explicit regrading, which may itself later be superseded.

1.3 Corpus census

The combined audit covered the repository and the larger SCI archive. It indexed 1,319 files, representing 791 distinct SHA-256 hashes. Of these, 491 unique text-bearing sources were extracted into the claim-and-coverage ledger; 155 were classified as substantive scientific sources and 132 as manuscript files. Exact duplicate groups and cross-format renderings were consolidated, while their paths were retained. No text extraction failed in the final index.

The 491-source crosswalk contains 148 implementation files, 126 manuscripts, 105 data or certificate files, 64 supporting texts, 23 orientation or ledger files, 19 manifests, and 6 historical manuscripts. The machine-readable ledger records every source's full digest, format, role, approximate word count, leading topics, claim labels, assigned volume and section, and review status.

Scientific grades and inference rules

2.1 A typed claim system

The inherited corpus frequently used `PROVED` for several logically different achievements. This edition replaces that single word with a typed system.

TABLE 2.1. Scientific grades used throughout the living edition.

Grade	Criterion
Definition	Fixes vocabulary or a mathematical object. It is neither true nor false until judged for consistency and usefulness.
Framework postulate	A declared starting commitment. Consequences remain conditional on it.
Exact identity	Follows by algebra from displayed definitions, with no physical identification implied.
Internal lemma/theorem	Follows from stated formal hypotheses by a checkable proof.
Computational result	Regenerated by a declared algorithm and certificate; scope is limited by finite precision, enumeration completeness, and implementation assumptions.
Established external result	A published theorem or observation imported with its actual hypotheses, model, units, and uncertainty.
Conditional identification	Connects an internal object with a physical one only after a named realisation map is granted.
Interpretation	A coherent reading that does not itself add a testable dynamical consequence.
Conjecture	A mathematical or physical statement for which the required proof or mechanism is absent.
Forecast	A predeclared output exposed to future or withheld data.
Open obligation	A missing bridge whose completion is required by a stronger claim.

Grade	Criterion
Withdrawn	A retained historical route that no longer supports its conclusion.

2.2 The claim tuple

Each serious claim should be representable as

$$\mathfrak{C} = (S, H, D, M, R, G, F), \quad (2.1)$$

where S is the statement, H its hypotheses, D the derivation or data, M the measurement map, R the reproducible artefact, G the grade, and F a falsifier or supersession condition. A purely mathematical identity may leave M empty. A physical claim may not.

The following invalid moves recur in speculative theory and are explicitly prohibited:

1. promoting an exact identity to a physical law without a realisation map;
2. treating numerical agreement after model selection as a forecast;
3. using a standard result as evidence for the framework merely because the framework can restate it;
4. counting correlated parameters as independent confirmations;
5. declaring a missing mechanism “forced” by a verbal definition;
6. changing conventions, datasets, stage counts, or branches after viewing the residual without recording the selection;
7. interpreting failure to falsify as confirmation.

2.3 Dependency and proof discipline

For each theorem, the proof graph must distinguish four edge types:

$$\begin{aligned} &\text{definition} \longrightarrow \text{formal consequence} \\ &\longrightarrow \text{realisation assumption} \longrightarrow \text{empirical readout.} \end{aligned} \quad (2.2)$$

An external theorem enters only at the edge licensed by its hypotheses. For example, lattice classification can identify E_8 after rank, integrality, evenness, unimodularity, and positive definiteness have been supplied; it cannot supply those premises. Gleason’s theorem constrains probability measures under non-contextual additivity hypotheses (Gleason, 1957; Busch, 2003; Caves et al., 2004); it does not derive those hypotheses from fairness language. Wigner and Stone

constrain transformations only after transition-probability preservation, bijectivity, and continuity have been stated (Wigner, 1959; Stone, 1932).

Commitments and definitions

3.1 Commitments A1–A6

The six inherited axioms are preserved below with their living-edition status.

TABLE 3.1. Foundational commitments.

ID	Name and grade	Scientific statement
A1	Systemhood; framework postulate	A model may treat a system as one entity while permitting internally distinguishable states or parts. This is a modelling convention, not an empirical law.
A2	Ground; ontological postulate	The first distinction is said to constitute its own “where”. No spacetime consequence follows until a mathematical state space and geometry are constructed.
A3	Two-sided reading; framework postulate	A mixed relation is counted in both orientations. This supplies the factor of two in the retained-fraction normalization; all downstream numerical claims remain conditional on it. It does not by itself prove evenness or lattice self-duality.
A4	Saturation; open physical postulate	A single accumulating channel is assumed to reach finite capacity. Capacity, threshold, state space, and dynamics remain unspecified.
A5	Conservation of readout; conservation postulate	Continuing state, outgoing record, and their order are assumed to conserve a declared total. Physical use requires an explicit conserved measure or current.

ID	Name and grade	Scientific statement
A6	Admissible reduction; definition	A reduction is declared unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions. Operator-algebra consequences require a specified algebra, action, norm, and fixed space.

3.2 Definitions D1–D18

TABLE 3.2. Framework vocabulary, preserved and delimited.

ID	Term	Living-edition definition
D1	Distinction	A relation or property that prevents complete equivalence under the comparison being made.
D2	Identity	Recoverability of declared distinctions through an admissible family of changes.
D3	Relation	A rule by which distinct contents compare, constrain, or transform one another without automatic identification.
D4	Seven doings	Distinguish, hold, close, answer, change, order, and record: the corpus’s constitutive grammar.
D5	Traversal stages	Persist, distinguish, relate, propagate, resolve, return, and close: a process-oriented rendering of D4.
D6	Object	A stable equivalence class of closed traversals relative to a declared observation and timescale.
D7	Update	The two-channel map $(a, b) \mapsto (a + b, a)$.
D8	Channels	The outward share u and retained share r , normalized by $u + r = 1$.
D9	Three weights	$W_L = u^2$, $W_B = 2ur$, and $W_M = r^2$. The physical names are interpretations.
D10	Crossing	One enacted step of self-description across a declared channel. Its assignment of cost r is a separate conjectural rule.

ID	Term	Living-edition definition
D11	Record	A distinction that remains discriminable and recoverable under the future dynamics relevant to the task. The inherited binary criterion is an idealization; physical records can be noisy or graded.
D12	Observer	A system in which another system's distinction becomes coupled and persistently recorded; a mechanical observer includes preparation, coupling, propagation, resolution, amplification, and record retention.
D13	Carrier	The structure on which the framework writes self-describing records. Self-duality is a modelling requirement, not a derivation of an even unimodular lattice.
D14	Primitive cross- ing	A smallest nonzero displacement in a chosen carrier model; in the E_8 realisation these are roots of squared norm two.
D15	Cluster projec- tion	A map C from carrier observables to a readable subalgebra satisfying A6.
D16	Decay generator	An operator K whose kernel is the readable algebra; the minimal choice is $I - C$, while a general positive generator may have a richer spectrum.
D17	Readable alge- bra	The subalgebra fixed by the declared reduction and therefore retained by the effective description. Its identification with physical observables is model-dependent.
D18	Geometry projectors	Orthogonal projectors P_X, P_T, P_I assigned to external distinction, causal order, and closure, with $P_X + P_T + P_I$ the geometry block. Their ranks are assumptions until represented independently.

Regraded formal register T1–T39

The following table is the controlling scientific register. It preserves the inherited numbering so every earlier proof, certificate, and cross-reference remains traceable. The short ruling is not a replacement for the fuller derivations in Volumes I–IV; it states the maximum defensible grade.

TABLE 4.1. Living-edition regrade of the inherited theorem register.

ID	Grade	Scientific ruling
T1	Internal finite-order lemma	Unique order follows if the seven dependency relations form the asserted chain; the partial order and uniqueness must be written explicitly.
T2	Internal uniqueness lemma	The Fibonacci update is unique only within the declared class of nonnegative copy-and-add matrices.
T3	Exact algebra	The positive projective fixed point of the stated recurrence is φ . Physical relevance is separate.
T4	Exact conditional algebra	$r = (2\varphi)^{-1}$ follows from the A3 normalization; the factor of two is assumed.
T5	Exact identity	$(u + r)^2 = u^2 + 2ur + r^2 = 1$. Sector names do not follow from the algebra.
T6	Conjectural quantization rule	Binary record status does not entail a physical crossing price exactly r . Preserve as a testable bridge.
T7	Exact conditional identity	Given r and three crossings, $r^3 = (\sqrt{5} - 2)/8$.
T8	Conditional classification	The rank-eight positive-definite even unimodular lattice is uniquely E_8 ; rank, integrality, evenness, unimodularity, and positivity must each be supplied (Conway and Sloane, 1999).
T9	Incomplete argument	Reading a relation twice does not prove an even integral bilinear form. This is an open target lemma.

ID	Grade	Scientific ruling
T10	Root-system lemma; conditional physics	Suitable E_8 roots close a triangle. Its status as the unique minimal physical history depends on admissible-move rules.
T11	Exact geometry, corrected	For roots of squared norm two with inner product -1 , parallelogram area is $\sqrt{3}$ and triangle area is $\sqrt{3}/2$.
T12	Reproducible computation	The Coxeter-plane radii exhibit the stated golden pairing under the recorded projection and tolerance; a symbolic proof would strengthen the result.
T13	Internal operator result	If $C^2 = C$, then $\ker(I - C) = \text{ran } C$. Calling the range physical is an identification.
T14	Exact operator observation	$I - C$ has a restricted spectrum; a general positive generator with the same kernel need not. No mass spectrum follows.
T15	Structural assumption	One-dimensional temporal order is not derived from the word “record”; rank-one time is an explicit premise.
T16	Structural assumption	One enacted closure does not prove a one-dimensional closure representation.
T17	Conditional metric construction	A declared three-plus-one split with opposite signs is Lorentzian. The ranks and signs require independent support.
T18	Established theorem, delimited	Alexandrov–Zeeman-type causal-preserver theorems recover Lorentz/conformal structure under dimensional, bijective, and causal hypotheses (Zeeman, 1964; Alexandrov, 1967).
T19	Withdrawn as proof	Lack of a distinguishing event does not force coherent quantum superposition; a classical mixture remains an alternative. A full operational reconstruction is required (Chiribella et al., 2011).
T20	Conditional representation	Continuous unit-modulus characters of additive action are exponentials. Complex Hilbert structure needs further composition and operational axioms.

ID	Grade	Scientific ruling
T21	Established theorem under hypotheses	Gleason/Busch/Caves-type results constrain noncontextual additive probability measures; fairness language alone is insufficient (Gleason, 1957; Busch, 2003; Caves et al., 2004).
T22	Established theorems under hypotheses	Wigner applies to transition-probability-preserving bijections and Stone to strongly continuous unitary groups; identifying the generator as energy is physical (Wigner, 1959; Stone, 1932).
T23	Exact algebra; conditional physics	The cross term is $2ur$. Universal interpretation as interference or boundary weight needs a model-specific amplitude map.
T24	Algebraic construction; conjectural gravity	A squared defect has three coefficient classes. A gravitational action also requires connection, curvature, invariant pairing, dimensions, symmetry breaking, and variation (MacDowell and Mansouri, 1977).
T25	Classification proposal	Commuting, sector-connecting, and noncommuting operations form a useful taxonomy; they do not derive gauge particles, matter, or masses.
T26	Persistence criterion	Retention rather than overwrite defines longitudinal identity within the framework; it is not a universal theorem forbidding replacement dynamics.
T27	Internal consequence plus interpretation	A coherent state is not a definite record under D11. Intrinsic unreadability of outcome selection remains interpretive.
T28	Physical conjecture	General relativity admits recollapsing cosmologies. A no-overwrite law must be realised covariantly before it can exclude them.
T29	Conditional rank bookkeeping	If the geometry block has rank five and time and closure each rank one, the remaining external rank is three.
T30	Finite Lie-theoretic result	The stated rank-three closed-root-subsystem classes may be retained with an exact classification or exhaustive certificate and a precise definition of subsystem.

ID	Grade	Scientific ruling
T31	Multi-stage conjectural selection	Complement calculations can be exact; irreducibility alone does not force one decay operator, and a 1 + 3 index split does not derive three particle families.
T32	Reproducible finite construction	Antipodal roots guarantee zero total sum; the claimed 240-step ordering additionally requires explicit adjacency conditions and an independently checked witness.
T33	Conjectural action identification	The coefficient pattern contacts MacDowell–Mansouri gravity, but term-level identity awaits the full differential-form action and field equations.
T34	Open; inherited proof rejected	One closure does not imply every loop contracts, and one world does not imply simple connectivity. Perelman’s theorem applies only after a closed simply connected three-manifold is established (Perelman, 2002).
T35	Established hydrogen mathematics; circular bridge	Fock’s S^3 follows from the Coulomb problem. Finite-design quadrature is valid, but importing that sphere cannot independently derive hydrogen (Fock, 1935 ; Delsarte et al., 1977).
T36	Conjectural chirality mechanism	A finite decay group alone does not provide a complete Lorentz-covariant chiral field theory or evade all E_8 representation constraints (Disler and Garibaldi, 2010).
T37	Conditional operator result plus finite search	Tomiyama and finite-group averaging apply under specified algebraic hypotheses; uniqueness and completeness of the root-generated search must be certified (Tomiyama, 1957).
T38	Standard GUT reproduction	The exterior-algebra charge table and 3/8 unification value are established $SU(5)/SO(10)$ representation theory. The traversal origin of the three-plus-two split remains unproved.
T39	Exact internal number; speculative map	The power of r is algebraically checkable. Its equality with the measured Planck/vacuum hierarchy requires a dimensionally complete, non-circular convention and uncertainty map.

4.1 Principal corrections

Several rulings materially change the inherited register.

1. T6 no longer treats binary record status as a derivation of an exact crossing cost.
2. T9 remains open because evenness requires an integral bilinear form, not a verbal two-sided reading.
3. T11 distinguishes the parallelogram area $\sqrt{3}$ from the triangular area $\sqrt{3}/2$.
4. T15 and T16 are structural rank assumptions, not proofs from singleness language.
5. T19's claimed derivation of superposition is withdrawn.
6. T28 is not a theorem of general relativity.
7. T34's route to S^3 is rejected because closure and uniqueness do not imply simple connectivity.
8. T35 retains the finite-design result but rejects the circular hydrogen derivation.
9. T36 does not use a finite group as a blanket escape from a field-theoretic no-go result.
10. T38 is credited to standard grand-unified representation theory.
11. T39 separates an exact internal number from its speculative physical map.

These are not cosmetic caveats. They determine what future work must calculate.

Reproducibility and computational certificates

5.1 Minimum reproducible unit

A numerical result is reproducible only when the following object exists:

$$\mathcal{U} = (I, C, E, O, V), \quad (5.1)$$

where I identifies immutable inputs, C identifies code, E records the execution environment, O contains outputs, and V independently verifies declared invariants. Each component must be content-addressed or versioned. A screenshot or rounded table is not a certificate.

For stochastic computation, I includes random seeds and the generator. For optimization or search, the certificate must separate discovery from verification. For exhaustive claims, the search space, pruning rules, symmetry quotient, and completion count must be explicit. For floating-point identities, tolerances and conditioning must be stated; symbolic or integer verification is preferred when available.

5.2 Certificate classes in the corpus

The reproducibility layer contains several distinct objects:

Algebra certificates

regenerate r, u , the three weights, closed forms, and cosmological dictionary values;

Root-system certificates

construct E_8 , projections, subgroup complements, ring radii, and candidate traversals;

Geometry certificates

expand the gravitational square and verify coefficient ratios;

Cosmology certificates

integrate background quantities, compute residual ledgers, and generate the master readout;

Data audits

verify the Atomic Web records and catalogue transformations;

Manifests

inventory paths and hashes but do not validate scientific conclusions.

5.3 Independent verification

The same implementation that discovers a witness should not be the sole verifier. For a root traversal, a small verifier should read the witness and check membership, uniqueness, adjacency, and closure without rerunning the search. For a cosmological table, a verifier should recompute formulas from a transparent parameter file and compare at full precision. For a document build, the source, bibliography, generated tables, compiler, and rendered pages are all part of the reproducible artefact.

EXPOSURE 6.1 — NUMERICAL PROOF BOUNDARIES

Machine-precision agreement is evidence for a computational statement, not automatically an exact theorem. An exhaustive result is only as complete as its enumeration and symmetry handling. The register must say whether a result is symbolic, integer-exact, interval-certified, exhaustively finite, or floating-point empirical.

5.4 Recommended release layout

Each public release should contain:

1. the five LaTeX sources and omnibus source;
2. the bibliography with persistent identifiers;
3. the full source and duplicate ledgers;
4. a machine-readable claim register;
5. data and code manifests with SHA-256 digests;
6. independent verification scripts;
7. build instructions and environment versions;
8. rendered PDFs plus page-image quality-control results;
9. a correction and supersession log.

Measurement, statistics, and falsification

6.1 The measurement dossier

Every empirical claim should ship with a dossier containing:

1. the physical mechanism and state variables;
2. the observable and its units;
3. the channel and propagation model;
4. instrument response, calibration, and selection function;
5. the estimator and full covariance;
6. nuisance parameters and priors;
7. the comparison model and likelihood;
8. preregistered acceptance, revision, and rejection rules;
9. data and code sufficient to reproduce the result where licensing permits.

This is the concrete implementation of the observational chain in Volume III.

6.2 Reconstruction is not prediction

The corpus contains several legitimate reconstructions: Standard Model charge branching, hydrogenic harmonic sampling, the MacDowell–Mansouri coefficient contact, and algebraic cosmological readouts. A reconstruction shows compatibility or compression. It becomes a prediction only if the model and analysis are fixed before the relevant target is used and if the result is sufficiently discriminating against alternatives.

For a dataset D , model M , nuisance parameters ν , and theory parameters θ , the comparison object is a likelihood or sampling distribution

$$p(D \mid \theta, \nu, M), \tag{6.1}$$

not a list of percentage differences. Correlated cosmological parameters require the joint covariance. A mean absolute pull calculated after selecting quantities

and transformations is descriptive unless selection is included in the inferential procedure.

6.3 Negative results belong in the main theory

The following negative results are scientifically productive and remain visible:

- vacuum crowding is too small to supply galactic dark matter in the tested form;
- geometry alone cannot separate α_L from ν in the Bubble compression kernel;
- family projections that give equal readouts do not explain observed family hierarchies;
- the Saha CMB solver places last scattering too early and does not calculate the observed spectra;
- the hydrogen sphere bridge is circular without an independently derived Coulomb interaction;
- the superposition, S^3 , chirality, and vacuum-hierarchy arguments do not presently reach their inherited grades;
- the cosmological master score is in-sample and does not possess a calibrated discovery significance.

6.4 Falsification must be model-specific

A framework this broad cannot be defeated by one vague mismatch, but each physical realisation can and should expose itself. A falsifier must bind a named version, observable map, parameter freedom, data domain, and threshold. It must also state what survives. Rejecting the dark-matter realisation would not invalidate the golden-ratio identity; rejecting the closure-to-e-fold map would not invalidate the finite root construction.

Historical consolidation and supersession

7.1 How the corpus developed

The source chronology can be summarized in five scientific phases:

1. **Foundational grammar:** distinction, identity, relation, the seven doings, the two-channel recurrence, and the three weights;
2. **Carrier construction:** E_8 , root geometry, clustering, decay, and the readable algebra;
3. **Traversal and observation:** helical lift, record order, quantum interfaces, Logical Action, Personal Relativity, and Universopedia;
4. **Physical applications:** gravity, particle branching, hydrogen, dark sectors, cosmological dictionaries, and empirical programmes;
5. **Audit and capstone:** formal registers, addenda, computational certificates, resolution ledgers, the Field Ledger, later matter/rate papers, and the Nine-Step Closure.

The five-volume architecture follows this dependency chronology rather than file creation time, because later documents frequently restate earlier material more clearly and occasionally correct it.

7.2 Consolidation rules

Exact duplicate files are represented once scientifically and many times in provenance. PDF and DOCX copies with the same intellectual content are treated as renderings of one source unless their text differs. Generated tables are not counted as independent evidence when they are deterministic outputs of the same script. Later restatements are merged into the strongest readable formulation while divergences enter the supersession ledger.

No information is discarded merely because it is scientifically weak. Instead it receives one of four dispositions:

$$\text{integrated,} \quad \text{historical,} \quad \text{superseded,} \quad \text{withdrawn.} \quad (7.1)$$

This prevents both inflation of the series by repetition and erasure of the research history.

7.3 Named supersessions

The principal supersessions include:

- “traversal payment as a universal one-third constant” becomes an audit-dependent typing score;
- “superposition forced by nonselection” becomes an operational postulate awaiting reconstruction;
- “three modes plus closure imply S^3 ” becomes an open topology problem;
- “hydrogen derived from carrier self-duality” becomes a finite-design quadrature result plus an open interaction bridge;
- “MacDowell–Mansouri term for term” is first narrowed to a coefficient contact and later completed as a registered action variation in the Nine-Step Closure;
- “finite decay evades the E_8 chirality obstruction” is replaced by the explicit chiral 16, its charge and anomaly certificates, and the separately localized microscopic index construction;
- “vacuum crowding supplies galactic dark matter” is withdrawn;
- every edition freeze becomes a historical analysis lock, not an immutable scientific state.

Coverage assurance and handoff

8.1 What complete coverage means

Complete coverage is an auditable relation between corpus sources and series dispositions, not an assertion that every duplicated paragraph appears verbatim. The machine-readable crosswalk supplied with this edition lists all 491 unique text-bearing sources. It is an archival mapping layer, distinct from the theorem and dependency registers printed in this volume. The full 1,319-file historical inventory and duplicate groups preserve non-text assets and alternate copies.

Coverage is checked at three levels:

Source level

every unique source has a volume, section, role, and disposition;

Claim level

each substantive mathematical or physical claim is integrated, regraded, superseded, withdrawn, or named as open;

Artefact level

code, data, certificates, images, manifests, and rendered copies remain traceable even when not printed in prose.

8.2 Closure reconciliation and active frontier

The nine-item queue originally printed here was a sound dependency order, but it is not the current status of the theory. The later closure volume formalizes the record calculus, consolidates the carrier hypotheses, regenerates all twenty-two corpus certificates, varies the local action, fixes the chiral matter and anomaly chain, writes the dark-sector effective tensors and perturbations, replaces Saha decoupling with CAMB recombination and Einstein–Boltzmann evolution, and freezes the observable maps and preregistration rules (Shields, 2026).

After that chronological reconciliation, the active frontier is localized:

1. solve the twelve-component stationary mass selector and connect its spec-

- trum to the absolute scale;
- 2. complete the discrete microscopic matter-to-cell constitutive identity;
- 3. derive the short-range granularity response kernel;
- 4. determine the carrier threshold spectrum and overall gauge normalization;
- 5. perform a precision public-likelihood analysis with an independent HyRec/CosmoRec cross-run;
- 6. execute the preregistered prospective physical measurements.

These entries are calculations, constitutive closures, and measurements at named interfaces. They do not reopen the already registered algebraic, geometric, action, matter, or cosmological dependency chain.

8.3 Final scientific position

Unified Mechanics is a theory of everything in the precise sense defined by this corpus: it proposes one generative structure beneath general relativity, quantum field theory, and quantum mechanics, recovers their successful domains through a common carrier-and-readout construction, and exposes the additional relations fixed by that construction to calculation and experiment. The scientific discipline of the theory is statement-specific: exact results remain exact, conditional realizations retain their premises, numerical outputs retain their certificates, and prospective measurements remain prospective. The remaining frontier is finite and named. It is the normal finishing work of a functioning theory, not an absence of theoretical architecture.

The complete source-disposition crosswalk is delivered as `audit/Unified_Mechanics_Source_Coverage_Ledger.csv`. Keeping the full table outside the printed volume preserves its audit function while leaving the formal scientific register readable.

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